

Alexander Yao

(908) 381-9450 alex.r.yao@gmail.com Raritan, NJ
GitHub LinkedIn Website

EDUCATION

University of Maryland, College Park
B.S. in Computer Science
Minor in Applied Mathematics

College Park, MD
Expected Dec. 2026

- **Relevant Coursework:** Algorithms, Compilers, Advanced Data Structures, Computer Systems, Programming Languages, Web Development, Parallel Computing, Probability and Statistics, Computer and Network Security

SKILLS

Languages: Python, Java, Rust, C, OCaml, JavaScript, SQL, Racket, CSS, HTML

Systems & Backend: Linux, Async I/O, QUIC

Tools: Git, Pandas, Playwright, Node.js, Microsoft Azure

EXPERIENCE

George Mason University
Computer Vision Research Assistant

Fairfax, VA
May 2025 – Present

- Designed a low-latency, cloud-native media delivery system over QUIC, achieving <150ms end-to-end latency.
- Built a scalable bidirectional client-server architecture integrating Python async I/O and Android clients for continuous media streams.
- Improved system reliability by 30% through packet scheduling, flow control tuning, and congestion simulation.
- Developed security testing modules to audit QUIC handshake and encryption behavior, strengthening transport-layer robustness.

University of Science and Technology of China
AI Security Research Intern

Hefei, China
June 2025 – August 2025

- Designed and executed controlled experiments simulating instruction-based and clean-label backdoor attacks against large language models.
- Built automated Python evaluation pipelines using statistical analysis to assess robustness and evasion behavior.
- Presented technical findings to an international academic audience of 200+ attendees.

PROJECTS

Personal Portfolio
React, TypeScript, Vite, CSS, Git

Jan. 2026

- Engineered a high-performance portfolio using React Router v7 and TypeScript, implementing a centralized design system with native light/dark mode support via CSS variables.
- Designed a responsive media gallery component that optimizes page load by lazy-loading embeds and providing custom loading states for unlisted video content.
- Built a mobile navigation system, utilizing dynamic animations to maintain UI consistency across various viewports.
- Automated the deployment pipeline using GitHub Actions to build and host the production application on GitHub Pages upon every main-branch push.

Cross-Platform Video Performance Tracker
Python, SQL, Pandas

Dec. 2025

- Designed a cross-platform data ingestion pipeline aggregating engagement metrics from YouTube and TikTok APIs.
- Implemented a headless browser scraping system using Playwright with session persistence and anti-bot mitigation to ensure reliable data collection.
- Developed platform-agnostic engagement normalization metrics (likes + comments per view) to enable fair, data-driven comparison across platforms.
- Transformed raw event data into clean, queryable analytics tables to support performance analysis.

Loot+ Systems Programming Project
Racket, x86-64 Assembly

May 2025

- Implemented a Racket-to-x86-64 compiler supporting closures, variadic functions, and dynamic arity checks.
- Built a custom exception-handling mechanism via stack unwinding and non-local jumps using low-level register manipulation.
- Implemented a full compiler pipeline (parsing, IR generation, codegen), achieving 100% behavioral parity with an interpreted reference model.